# SUMS OF POWERS VIA CENTRAL FINITE DIFFERENCES AND NEWTON'S FORMULA

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Abstract. In this manuscript, we derive closed formulas for multifold sums of powers of integers by combining the central Newton interpolation formula with hockey-stick identities for binomial coefficients. We further provide Wolfram Mathematica programs for the efficient verification of the derived identities.

## Contents



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## 1. Auxiliary lemmas

In this section, allow us to define a few lemmas, definitions, and conventions that serve as foundation for main results of this manuscript. For multifold sums of powers, we use the notation provided by Donald Knuth in [1], because it is simple and beautiful

**Definition 1.1** (Multifold sums of powers). *For non-negative integers $r, n, m$*

$$\begin{aligned}
\Sigma^0 n^m &= n^m, \\
\Sigma^1 n^m &= \Sigma^0 1^m + \Sigma^0 2^m + \cdots + \Sigma^0 n^m, \\
\Sigma^{r+1} n^m &= \Sigma^r 1^m + \Sigma^r 2^m + \cdots + \Sigma^r n^m.
\end{aligned}$$

We utilize the following notation for falling factorials

**Definition 1.2** (Falling factorials). *For integers $x, n$*

$$(n)_k = \begin{cases} 1, & \text{if } k = 0, \\ n(n-1)(n-2)\cdots(n-k+1) = \prod_{j=0}^{k-1}(n-j), & \text{if } k > 0, \\ 0, & \text{else.} \end{cases}$$

We utilize the following notation for central factorials

**Definition 1.3** (Central factorials). *For integers $n, k$*

$$n^{[k]} = \begin{cases} 0, & \text{if } k < 0, \\ 1, & \text{if } k = 0, \\ n\left(n+\frac{k}{2}-1\right)\left(n+\frac{k}{2}-2\right)\cdots\left(n-\frac{k}{2}+1\right) = n\left(n+\frac{k}{2}-1\right)_{k-1}, & \text{if } k > 0, \end{cases}$$

*where $\left(n+\frac{k}{2}-1\right)_{k-1} = \prod_{j=0}^{k-2}\left(n+\frac{k}{2}-1-j\right)$ are falling factorials.*

We encourage the audience to read J. F. Steffensen's work on the definition of generalized factorials [2].

**Lemma 1.4** (Central Newton's formula)**.** *For a function $f$ defined on integers and arbitrary integers $x, t$*

$$f(x) = \sum_{k=0}^{\infty} \frac{(x-t)^{[k]}}{k!} \delta^k f(t),$$

*where $\delta^k f(t)$ denotes the $k$-th central finite difference*

$$\delta^k f(t) = \sum_{j=0}^{k} (-1)^j \binom{k}{j} f\left(t + \tfrac{k}{2} - j\right),$$

*and $(x-t)^{[k]}$ are central factorials defined by (1.3).*

*Proof.* See [3, p. 462], and [4, section 2, eq. (19)], and [5, section 6, eq. (21)]. □

We have

**Lemma 1.5** (Newton's formula for powers)**.** *For non-negative integers $n, m$, and an arbitrary integer $t$,*

$$n^m = \sum_{k=0}^{m} \frac{(n-t)^{[k]}}{k!} \delta^k t^m.$$

*Proof.* Special case of Lemma (1.4) for $f(n) = n^m$. □

We have

**Lemma 1.6** (Central factorials binomial form)**.** *For integers $n$ and $k \neq 0$,*

$$\frac{n^{[k]}}{k!} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1}.$$

*Proof.* We have

$$\frac{n^{[k]}}{k!} = \frac{n}{k!} \left(n + \frac{k}{2} - 1\right)_{k-1} = \frac{n}{k(k-1)!} \left(n + \frac{k}{2} - 1\right)_{k-1} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1},$$

because of the identity in falling factorials $\frac{(x)_n}{n!} = \binom{x}{n}$. □

We have

**Lemma 1.7** (Generalized hockey-stick identity)**.** *For arbitrary integers $a, b$ and $j$,*

$$\sum_{k=a}^{b} \binom{k}{j} = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

*Proof.* We have, $\sum_{k=a}^{b} \binom{k}{j} = \binom{a}{j} + \binom{a+1}{j} + \cdots + \binom{b}{j} = \left(\sum_{k=0}^{b} \binom{k}{j}\right) - \left(\sum_{k=0}^{a-1} \binom{k}{j}\right)$. By hockey-stick identity $\sum_{k=0}^{n} \binom{k}{j} = \binom{n+1}{j+1}$ yields,

$$\sum_{k=a}^{b} \binom{k}{j} = \left(\sum_{k=0}^{b} \binom{k}{j}\right) - \left(\sum_{k=0}^{a-1} \binom{k}{j}\right) = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

This completes the proof. □

We have

**Lemma 1.8** (Centered hockey-stick identity)**.** *For integer $n \geq 1$, and arbitrary integers $t, k, r$*

$$\sum_{j=1}^{n} \binom{j - t + \frac{k}{2} + r}{k} = \binom{n - t + \frac{k}{2} + r + 1}{k+1} - \binom{1 - t + \frac{k}{2} + r}{k+1}.$$

*Proof.* By setting $a = 1 - t + \frac{k}{2} + r$, and $b = n - t + \frac{k}{2} + r$ into Lemma (1.7) yields

$$\sum_{j=1}^{n} \binom{j-t+\frac{k}{2}+r}{k} = \sum_{j=1-t+\frac{k}{2}+r}^{n-t+\frac{k}{2}+r} \binom{j}{k} = \binom{n-t+\frac{k}{2}+r+1}{k+1} - \binom{1-t+\frac{k}{2}+r}{k+1}.$$

This completes the proof. □

We have

**Lemma 1.9** (Multifold sums of zero powers)**.** *For non-negative integers $r, n$*

$$\Sigma^r \, n^0 = \binom{r+n-1}{r}.$$

*Proof.*

(1) Let $r = 0$, then we have $\Sigma^0 \, n^0 = 1$, by definition (1.1).

(2) Let $r = 1$, then we have $\Sigma^1 \, n^0 = \sum_{k=1}^{n} 1 = \binom{n}{1}$.

(3) Let $r = 2$, then we have $\Sigma^2 \, n^0 = \sum_{k=1}^{n} \binom{k}{1} = \binom{n+1}{2}$.

(4) Let $r = 3$, then we have $\Sigma^3 \, n^0 = \sum_{k=1}^{n} \binom{k+1}{2} = \binom{n+2}{3}$.

(5) Similarly, for arbitrary integer $r$, by hockey-stick identity $\sum_{k=1}^{n}\binom{k}{r}=\binom{n+1}{r+1}$, yields $\Sigma^r\, n^0=\sum_{k=1}^{n}\Sigma^{r-1}\, k^0=\sum_{k=1}^{n}\binom{k+r-2}{r-1}=\binom{r+n-1}{r}$.

This completes the proof. □

We follow the convention

**Convention 1.10.** *For all $x$*

$$x^0=1.$$

Donald Knuth give extensive discussion on the convention $x^0=1$ for all $x$, including zero, in Concrete Mathematics [6, p. 162], and in [7].

## 2. Introduction

In this manuscript, we derive formulas for sums of powers by combining central Newton's interpolation formula for powers (1.5) with hockey-stick identity (1.8). We implement the algorithm for finding closed forms of sums of powers of integers, proposed in [8, Alg. (11.2)]. More precisely, the main idea is to determine binomial basis of Newton's formula, in terms of variants of difference operators, for example

$$\begin{cases} f(x) &= \sum_{k=0}^{\infty}\frac{(x-a)^{[k]}}{k!}\delta^k f(a)=\sum_{k=0}^{\infty}\frac{x-a}{k}\binom{x-a+\frac{k}{2}-1}{k-1}\delta^k f(a),\\ f(x) &= \sum_{k=0}^{\infty}\frac{(x-a)_k}{k!}\Delta^k f(a)=\sum_{k=0}^{\infty}\binom{x-a}{k}\Delta^k f(a),\\ f(x) &= \sum_{k=0}^{\infty}\frac{(x-a)^{(k)}}{k!}\nabla^k f(a)=\sum_{k=0}^{\infty}\binom{x-a+k-1}{k}\nabla^k f(a),\\ f(x) &= \sum_{k=0}^{\infty}\frac{(x-a)^k}{k!}\frac{\mathrm{d}^k f(a)}{\mathrm{d}x^k}. \end{cases}$$

We may observe that each variation of Newton's formula above has its binomial basis

$$\begin{cases} \frac{(x)_n}{n!} &= \frac{1}{n!}x(x-1)(x-2)\cdots(x-n+1)=\binom{x}{n},\\ \frac{x^{(n)}}{n!} &= \frac{1}{n!}x(x+1)(x+2)\cdots(x+n-1)=\binom{x+n-1}{n},\\ \frac{x^{[n]}}{n!} &= \frac{1}{n!}\left(n+\frac{k}{2}-1\right)\left(n+\frac{k}{2}-2\right)\cdots\left(n-\frac{k}{2}+1\right)=\frac{x}{n}\binom{x+\frac{n}{2}-1}{n-1}. \end{cases}$$

This allows us to successfully combine Newton's formula with hockey-stick type identity, to find closed forms for sums of powers of integers. Thus, we focus on closed forms of powers sums, by utilizing central Newton's formula, and hockey-stick identity (1.8).

## 3. Main results

We start our discussion from Lemma (1.5), which yields backward Newton's interpolation formula for powers. Therefore, by applying Lemma (1.6) to Newton's interpolation formula (1.5), for non-negative integers $n, m$ and an arbitrary integer $t$ yields

$$\begin{aligned} n^m &= \frac{(n-t)^{[0]}}{0!}\delta^0 t^m + \sum_{k=1}^{m} \frac{n-t}{k}\binom{n+t+\frac{k}{2}-1}{k-1}\delta^k t^m \\ &= t^m + \sum_{k=1}^{m} (n-t)\binom{n-t+\frac{k}{2}-1}{k-1}\frac{\delta^k t^m}{k}. \end{aligned}$$

Note that we isolate the term $\frac{(n-t)^{[0]}}{0!}\delta^0 t^m$ to avoid division by zero. By expanding the brackets, we have

$$n^m = t^m + \sum_{k=1}^{m} \frac{\delta^k t^m}{k}\left[n\binom{n-t+\frac{k}{2}-1}{k-1} - t\binom{n-t+\frac{k}{2}-1}{k-1}\right].$$

Now we notice that sum above has an additional factor $n$ in its binomial basis $n\binom{n-t+\frac{k}{2}-1}{k-1}$, preventing us from using centered hockey-stick identity (1.8) for closed form. Thus, consider the following Lemma

**Lemma 3.1** (Binomial decomposition)**.** *For non-negative integers $r, n, m$*

$$n\binom{n+r}{m} = (m+1)\binom{n+r}{m+1} - (r-m)\binom{n+r}{m}.$$

*Proof.* By expanding the brackets yields,

$$n\binom{n+r}{m} = m\binom{n+r}{m+1} + \binom{n+r}{m+1} - r\binom{n+r}{m} + m\binom{n+r}{m}.$$

Recall the extraction property of binomial coefficients $\binom{n}{k+1} = \frac{n-k}{k+1}\binom{n}{k}$. Therefore, by extraction property

$$\binom{n+r}{m+1} = \frac{n+r-m}{m+1}\binom{n+r}{m}.$$

Thus,

$$n\binom{n+r}{m} = m\frac{n+r-m}{m+1}\binom{n+r}{m} + \frac{n+r-m}{m+1}\binom{n+r}{m} - r\binom{n+r}{m} + m\binom{n+r}{m}.$$

By factoring out binomial coefficient $\binom{n+r}{m}$ yields

$$\begin{aligned}
n\binom{n+r}{m} &= \binom{n+r}{m}\left[m\frac{n+r-m}{m+1} + \frac{n+r-m}{m+1} - r + m\right] \\
&= \binom{n+r}{m}\left[(m+1)\frac{n+r-m}{m+1} - r + m\right] \\
&= \binom{n+r}{m}n.
\end{aligned}$$

This completes the proof. $\square$

Hence, by setting $r \to -t + \frac{k}{2} - 1$, and $m \to k-1$ into Lemma (3.1) gives

$$n\binom{n-t+\frac{k}{2}-1}{k-1} = k\binom{n-t+\frac{k}{2}-1}{k} + \left[t+\frac{k}{2}\right]\binom{n-t+\frac{k}{2}-1}{k-1}.$$

Because, by binomial decomposition (3.1), we have

$$\begin{aligned}
n\tbinom{n-t+\frac{k}{2}-1}{k-1} &= k\tbinom{n-t+\frac{k}{2}-1}{k-1+1} - \left[-t+\tfrac{k}{2}-1-(k-1)\right]\tbinom{n-t+\frac{k}{2}-1}{k-1} \\
&= k\tbinom{n-t+\frac{k}{2}-1}{k} - \left[-t-\tfrac{k}{2}\right]\tbinom{n-t+\frac{k}{2}-1}{k-1} \\
&= k\tbinom{n-t+\frac{k}{2}-1}{k} + \left[t+\tfrac{k}{2}\right]\tbinom{n-t+\frac{k}{2}-1}{k-1}.
\end{aligned}$$

Thus,

$$n^m = t^m + \sum_{k=1}^{m}\frac{\delta^k t^m}{k}\left[k\binom{n-t+\frac{k}{2}-1}{k} + \left[t+\frac{k}{2}\right]\binom{n-t+\frac{k}{2}-1}{k-1} - t\binom{n-t+\frac{k}{2}-1}{k-1}\right].$$

By collapsing the terms $t\binom{n-t+\frac{k}{2}-1}{k-1}$, and by simplifying the factors $k$ yields

$$n^m = t^m + \sum_{k=1}^{m}\left[\binom{n-t+\frac{k}{2}-1}{k} + \frac{1}{2}\binom{n-t+\frac{k}{2}-1}{k-1}\right]\delta^k t^m.$$

We move the term $t^m$ under the summation, thus

$$n^m = \sum_{k=0}^{m}\left[\binom{n-t+\frac{k}{2}-1}{k} + \frac{1}{2}\binom{n-t+\frac{k}{2}-1}{k-1}\right]\delta^k t^m.$$

Now, we simplify the binomial basis $\binom{n-t+\frac{k}{2}-1}{k}+\frac{1}{2}\binom{n-t+\frac{k}{2}-1}{k-1}$ to get rid of fractions inside of it. Let $a=n-t+\frac{k}{2}-1$, then

$$\binom{a}{k}+\frac{1}{2}\binom{a}{k-1}=\frac{1}{2}\left[2\binom{a}{k}+\binom{a}{k-1}\right]=\frac{1}{2}\left[\binom{a}{k}+\binom{a+1}{k}\right].$$

Therefore,

$$n^m=\frac{1}{2}\sum_{k=0}^{m}\left[\binom{n-t+\frac{k}{2}-1}{k}+\binom{n-t+\frac{k}{2}}{k-1}\right]\delta^k t^m.$$

Note that we do not bother ourselves about fractions in upper index of binomial coefficients $\binom{n-t+\frac{k}{2}-1}{k}$, and $\binom{n-t+\frac{k}{2}}{k-1}$, because binomial coefficient $\binom{n}{k}$ is a polynomial in $n$. The well-known identity in unsigned Stirling numbers of the first kind $\left[{n \atop k}\right]$ shows it explicitly

$$\binom{n}{k}=\sum_{j=0}^{k}\frac{(-1)^{k-j}}{k!}\left[{k \atop j}\right]n^j.$$

For example,

**Example 3.2.** *For integer $0\leq t\leq 3$, we have*

$$\begin{aligned}
t=0: \quad n^3 &= \left(1+\tbinom{n}{-1}\right)\cdot 0+\tfrac{1}{2}(1+2n)\cdot\tfrac{1}{4}+\tfrac{1}{2}(2+n+n^2)\cdot 0\\
&\quad+\tfrac{1}{48}(21+46n+12n^2+8n^3)\cdot 6,\\
t=1: \quad n^3 &= \left(1+\tbinom{n-1}{-1}\right)\cdot 1+\tfrac{1}{2}(-1+2n)\cdot\tfrac{13}{4}+\tfrac{1}{2}(2-n+n^2)\cdot 6\\
&\quad+\tfrac{1}{48}(-21+46n-12n^2+8n^3)\cdot 6,\\
t=2: \quad n^3 &= \left(1+\tbinom{n-2}{-1}\right)\cdot 8+\tfrac{1}{2}(-3+2n)\cdot\tfrac{49}{4}+\tfrac{1}{2}(4-3n+n^2)\cdot 12\\
&\quad+\tfrac{1}{48}(-87+94n-36n^2+8n^3)\cdot 6,\\
t=3: \quad n^3 &= \left(1+\tbinom{n-3}{-1}\right)\cdot 27+\tfrac{1}{2}(-5+2n)\cdot\tfrac{109}{4}+\tfrac{1}{2}(8-5n+n^2)\cdot 18\\
&\quad+\tfrac{1}{48}(-225+190n-60n^2+8n^3)\cdot 6.
\end{aligned}$$

Hence, we can utilize centered hockey-stick identity (1.8) for closed formula for sums of powers, because

$$\Sigma^1 n^m = \frac{1}{2}\sum_{k=0}^{m}\left[\sum_{j=1}^{n}\binom{j-t+\frac{k}{2}-1}{k} + \sum_{j=1}^{n}\binom{j-t+\frac{k}{2}}{k-1}\right]\delta^k t^m.$$

Therefore, by Lemma (1.8)

$$\begin{cases} \sum_{j=1}^{n}\binom{j-t+\frac{k}{2}-1}{k} & = \binom{n-t+\frac{k}{2}+0}{k+1} - \binom{0-t+\frac{k}{2}}{k+1}, \\ \sum_{j=1}^{n}\binom{j-t+\frac{k}{2}}{k-1} & = \binom{n-t+\frac{k}{2}+1}{k+1} - \binom{1-t+\frac{k}{2}}{k+1}. \end{cases}$$

Thus, closed formula for ordinary sums of powers follows

**Proposition 3.3** (Ordinary sums of powers)**.** *For non-negative integers $n, m$, and an arbitrary integer $t$*

$$\Sigma^1 n^m = \frac{1}{2}\sum_{k=0}^{m}\left[\binom{n-t+\frac{k}{2}+0}{k+1} + \binom{n-t+\frac{k}{2}+1}{k+1} - \binom{0-t+\frac{k}{2}}{k+1} - \binom{1-t+\frac{k}{2}}{k+1}\right]\delta^k t^m.$$

For example,

**Example 3.4.** *For integer $0 \leq t \leq 4$, we have*

$$\begin{aligned}
t=0: \quad \Sigma^1 n^3 &= 2n \cdot 0 + (n+n^2)\cdot \tfrac{1}{4} + \tfrac{1}{6}(n+3n^2+2n^3)\cdot 0 \\
&\quad + \tfrac{1}{24}(-n+n^2+4n^3+2n^4)\cdot 6, \\
t=1: \quad \Sigma^1 n^3 &= 2n \cdot 1 + (-n+n^2)\cdot \tfrac{13}{4} + \tfrac{1}{6}(n-3n^2+2n^3)\cdot 6 \\
&\quad + \tfrac{1}{24}(n+n^2-4n^3+2n^4)\cdot 6, \\
t=2: \quad \Sigma^1 n^3 &= 2n \cdot 8 + (-3n+n^2)\cdot \tfrac{49}{4} + \tfrac{1}{6}(13n-9n^2+2n^3)\cdot 12 \\
&\quad + \tfrac{1}{24}(-21n+25n^2-12n^3+2n^4)\cdot 6, \\
t=3: \quad \Sigma^1 n^3 &= 2n \cdot 27 + (-5n+n^2)\cdot \tfrac{109}{4} + \tfrac{1}{6}(37n-15n^2+2n^3)\cdot 18 \\
&\quad + \tfrac{1}{24}(-115n+73n^2-20n^3+2n^4)\cdot 6, \\
t=4: \quad \Sigma^1 n^3 &= 2n \cdot 64 + (-7n+n^2)\cdot \tfrac{193}{4} + \tfrac{1}{6}(73n-21n^2+2n^3)\cdot 24 \\
&\quad + \tfrac{1}{24}(-329n+145n^2-28n^3+2n^4)\cdot 6.
\end{aligned}$$

Similarly, by setting $r=0$, and $r=1$ into Lemma (1.8), we have

$$\begin{cases} \sum_{j=1}^{n}\binom{j-t+\frac{k}{2}+0}{k+1} &= \binom{n-t+\frac{k}{2}+1}{k+2}-\binom{1-t+\frac{k}{2}}{k+2}, \\ \sum_{j=1}^{n}\binom{j-t+\frac{k}{2}+1}{k+1} &= \binom{n-t+\frac{k}{2}+2}{k+2}-\binom{2-t+\frac{k}{2}}{k+2}. \end{cases}$$

Hence,

**Proposition 3.5** (Double sums of powers)**.** *For non-negative integers $n, m$, and an arbitrary integer $t$*

$$\begin{aligned} \Sigma^2 n^m = \frac{1}{2}\sum_{k=0}^{m}\Bigg[&\binom{n-t+\frac{k}{2}+1}{k+2}+\binom{n-t+\frac{k}{2}+2}{k+2} \\ &-\left(\binom{0-t+\frac{k}{2}}{k+1}+\binom{1-t+\frac{k}{2}}{k+1}\right)\Sigma^1 n^0 \\ &-\left(\binom{1-t+\frac{k}{2}}{k+2}+\binom{2-t+\frac{k}{2}}{k+2}\right)\Sigma^0 n^0\Bigg]\delta^k t^m. \end{aligned}$$

For example,

**Example 3.6.** *For integer $0 \le t \le 2$, we have*

$$\begin{aligned} t=0: \quad \Sigma^1 n^3 &= \tfrac{1}{2}(n+n^2)\cdot 0+\tfrac{1}{48}(-3-2n+12n^2+8n^3)\cdot\tfrac{1}{4} \\ &+\tfrac{1}{24}(-2n-n^2+2n^3+n^4)\cdot 0 \\ &+\tfrac{1}{3840}(45+18n-200n^2-80n^3+80n^4+32n^5)\cdot 6, \\ t=1: \quad \Sigma^1 n^3 &= \tfrac{1}{2}(-n+n^2)\cdot 1+\tfrac{1}{48}(3-2n-12n^2+8n^3)\cdot\tfrac{13}{4} \\ &+\tfrac{1}{24}(2n-n^2-2n^3+n^4)\cdot 6 \\ &+\tfrac{1}{3840}(-45+18n+200n^2-80n^3-80n^4+32n^5)\cdot 6, \\ t=2: \quad \Sigma^1 n^3 &= \tfrac{1}{2}(2-3n+n^2)\cdot 8+\tfrac{1}{48}(-15+46n-36n^2+8n^3)\cdot\tfrac{49}{4} \\ &+\tfrac{1}{24}(-6n+11n^2-6n^3+n^4)\cdot 12 \\ &+\tfrac{1}{3840}(105-142n-360n^2+560n^3-240n^4+32n^5)\cdot 6. \end{aligned}$$

Thus, in general

**Theorem 3.7** (Multifold sums of powers)**.** *For non-negative integers $r, n, m$, and an arbitrary integer $t$*

$$\begin{aligned}\Sigma^r n^m = \frac{1}{2}\sum_{k=0}^{m}\Bigg[&\binom{n-t+\frac{k}{2}+r-1}{k+r}+\binom{n-t+\frac{k}{2}+r}{k+r}\\ &-\sum_{s=0}^{r-1}\left(\binom{s-t+\frac{k}{2}}{k+s+1}+\binom{s-t+\frac{k}{2}+1}{k+s+1}\right)\Sigma^{r-1-s} n^0\Bigg]\delta^k t^m.\end{aligned}$$

By Lemma (1.9) pure binomial form follows

**Proposition 3.8** (Multifold Binomial sums of powers)**.** *For non-negative integers $r, n, m$, and an arbitrary integer $t$*

$$\begin{aligned}\Sigma^r n^m = \frac{1}{2}\sum_{k=0}^{m}\Bigg[&\binom{n-t+\frac{k}{2}+r-1}{k+r}+\binom{n-t+\frac{k}{2}+r}{k+r}\\ &-\sum_{s=0}^{r-1}\left(\binom{s-t+\frac{k}{2}}{k+s+1}+\binom{s-t+\frac{k}{2}+1}{k+s+1}\right)\binom{r-s+n-2}{r-s-1}\Bigg]\delta^k t^m.\end{aligned}$$

Therefore, we have successfully derived multifold formulas for sums of powers, in closed form. The work [9] provides implementation of algorithm proposed in [8, Alg. (11.2)], in terms of forward finite differences $\Delta$, and Newton's formula. Similarly, the work [10] discusses multifold formulas for sums of powers in terms of backward differences $\nabla$, and Newton's formula.

## 4. Remark on Knuth's formula

In [1] Donald Knuth provides a remarkable formula for sums of odd powers, in terms of central factorial numbers of the second kind $T(n, k)$

**Proposition 4.1** (Knuth's odd powers sum)**.** *For integers $n \geq 0$, and $m \geq 1$*

$$\Sigma^1 n^{2m-1} = \sum_{k=1}^{m}(2k-1)!\binom{n+k}{2k}T(2m, 2k),$$

*where $T(2m, 2k)$ are central factorial numbers of the second kind.*

See the sequences A008957, and A036969, and A269945 in the OEIS [11] for central factorial numbers of the second kind $T(n,k)$. For example,

| $n/k$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|
| 0 | 1 | | | | | | | |
| 1 | 0 | 1 | | | | | | |
| 2 | 0 | 1 | 1 | | | | | |
| 3 | 0 | 1 | 5 | 1 | | | | |
| 4 | 0 | 1 | 21 | 14 | 1 | | | |
| 5 | 0 | 1 | 85 | 147 | 30 | 1 | | |
| 6 | 0 | 1 | 341 | 1408 | 627 | 55 | 1 | |
| 7 | 0 | 1 | 1365 | 13013 | 11440 | 2002 | 91 | 1 |

**Table 1.** Central factorial numbers of the second kind $T(2n,2k)$.

It is quite interesting to notice that Proposition (4.1) originates from centered Newton's formula for powers (1.5) evaluated at zero. By central Newton's formula, for $m \geq 1$, we have

$$n^{2m} = \sum_{k=1}^{2m} \frac{n^{[k]}}{k!} \cdot \delta^k 0^{2m}$$

The iteration for $k=0$ can be omitted because $\delta^0 0^{2m} = 0$, for $m \geq 1$. Now, we observe that central difference $\delta^k 0^m$ satisfies the parity of arguments $m$ and $k$, such that

$$\begin{cases} \delta^k 0^m \neq 0, & \text{when} \quad m \equiv k \pmod 2, \\ \delta^k 0^m = 0, & \text{when} \quad m \not\equiv k \pmod 2, \end{cases}$$

meaning that $\delta^k 0^m$ is zero whether $k, m$ are odd and even, and vise versa. Note that the parity property holds only for central difference of powers evaluated at zero, by contrast $\delta^3 1^4 = 24 \neq 0$. Thus, the parity of $\delta^k 0^{2m}$ allows us to omit odd iterations of $k$, because $\delta^k 0^{2m}$ is zero for odd $k$

$$n^{2m} = \sum_{k=1}^{m} \frac{n^{[2k]}}{(2k)!} \cdot \delta^{2k} 0^{2m}.$$

By definition of central factorial numbers of the second kind, yields

$$n^{2m} = \sum_{k=1}^{m} n^{[2k]} T(2m, 2k).$$

Thus,

$$n^{2m} = \sum_{k=1}^{m} (2k)! \frac{n^{[2k]}}{(2k)!} T(2m, 2k)$$

By simplifying $\frac{n^{[2k]}}{(2k)!}$ using Lemma (1.6), we get

$$n^{2m} = \sum_{k=1}^{m} (2k)! \frac{n}{2k} \binom{n+k-1}{2k-1} T(2m, 2k).$$

Therefore,

$$n^{2m} = \sum_{k=1}^{m} (2k-1)! n \binom{n+k-1}{2k-1} T(2m, 2k).$$

By factoring out $n$, we obtain

$$n^{2m-1} = \sum_{k=1}^{m} (2k-1)! \binom{n+k-1}{2k-1} T(2m, 2k).$$

By using hockey-stick identity $\sum_{k=0}^{n} \binom{k}{j} = \binom{n+1}{j+1}$ yields

$$\Sigma^1 n^{2m-1} = \sum_{k=1}^{m} (2k-1)! \binom{n+k}{2k} T(2m, 2k).$$

The formula above matches the Proposition (4.1) precisely. Clearly, the formula for sums of odd powers can be generalized to multifold case, by applying hockey-stick identity $\sum_{k=0}^{n} \binom{k}{j} = \binom{n+1}{j+1}$ repeatedly

**Proposition 4.2** (Knuth's odd powers sum multifold)**.** *For non-negative integers $r, n,$ and $m \geq 1$*

$$\Sigma^{r+1} n^{2m-1} = \sum_{k=1}^{m} (2k-1)! \binom{n+k+r}{2k+r} T(2m, 2k),$$

*where $T(2m, 2k)$ are central factorial numbers of the second kind.*

For example,

**Example 4.3.** *For non-negative integers $n$, and $r = 1$, we have*

$$\Sigma^1 n^1 = \tbinom{n+1}{2}1,$$

$$\Sigma^1 n^3 = \tbinom{n+2}{4}6 + \tbinom{n+1}{2}1,$$

$$\Sigma^1 n^5 = \tbinom{n+3}{6}120 + \tbinom{n+2}{4}30 + \tbinom{n+1}{2}1,$$

$$\Sigma^1 n^7 = \tbinom{n+4}{8}5040 + \tbinom{n+3}{6}1680 + \tbinom{n+2}{4}126 + \tbinom{n+1}{2}1.$$

For example,

**Example 4.4.** *For non-negative integers $r, n$*

$$\Sigma^r n^1 = \tbinom{n+0+r}{1+r}1,$$

$$\Sigma^r n^3 = \tbinom{n+1+r}{3+r}6 + \tbinom{n+0+r}{1+r}1,$$

$$\Sigma^r n^5 = \tbinom{n+2+r}{5+r}120 + \tbinom{n+1+r}{3+r}30 + \tbinom{n+0+r}{1+r}1,$$

$$\Sigma^r n^7 = \tbinom{n+3+r}{7+r}5040 + \tbinom{n+3+r}{5+r}1680 + \tbinom{n+1+r}{3+r}126 + \tbinom{n+0+r}{1+r}1.$$

The coefficients $1, 6, 1, 120, 30, 1, 5040, 1680, \ldots$ is the sequence A303675 in the OEIS [11]. However, Proposition (4.2) serves as the source of inspiration for one more remarkable identity. By Newton's formula (1.5), and by Lemma (1.6), we have

$$n^m = \sum_{k=1}^{m} \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1} \delta^k 0^m$$

By factoring out $n$

$$n^{m-1} = \sum_{k=1}^{m} \frac{1}{k} \binom{n + \frac{k}{2} - 1}{k-1} \delta^k 0^m$$

By shifting

$$n^m = \sum_{k=1}^{m+1} \frac{1}{k} \binom{n + \frac{k}{2} - 1}{k-1} \delta^k 0^{m+1}$$

By re-indexing

$$n^m = \sum_{k=0}^{m} \frac{1}{k+1} \binom{n + \frac{k}{2} - \frac{1}{2}}{k} \delta^{k+1} 0^{m+1} \tag{1}$$

Thus, by setting $r = 0$ and $t = \frac{1}{2}$ into Lemma (1.8) yields

**Proposition 4.5** (Ordinary sums of powers at zero). *For non-negative integers $n, m$*

$$\Sigma^1 n^m = \sum_{k=0}^{m} \frac{1}{k+1} \left[ \binom{n + \frac{k}{2} + \frac{1}{2}}{k+1} - \binom{\frac{k}{2} + \frac{1}{2}}{k+1} \right] \delta^{k+1} 0^{m+1}.$$

Which can also be rewritten in terms of central factorial numbers of the second kind

$$\Sigma^1 n^m = \sum_{k=0}^{m} k! \left[ \binom{n + \frac{k}{2} + \frac{1}{2}}{k+1} - \binom{\frac{k}{2} + \frac{1}{2}}{k+1} \right] T(k+1, m+1).$$

By setting $r = 1$ and $t = \frac{1}{2}$ into Lemma (1.8), we get formula for double sums of powers

**Proposition 4.6** (Double sums of powers at zero). *For non-negative integers $n, m$*

$$\Sigma^2 n^m = \sum_{k=0}^{m} \frac{1}{k+1} \left[ \binom{n + \frac{k}{2} + \frac{3}{2}}{k+2} - \binom{\frac{k}{2} + \frac{1}{2}}{k+1} \Sigma^1 n^0 - \binom{\frac{k}{2} + \frac{3}{2}}{k+2} \Sigma^0 n^0 \right] \delta^{k+1} 0^{m+1}.$$

In general, we have

**Theorem 4.7** (Multifold sums of powers at zero). *For non-negative integers $r, n, m$*

$$\Sigma^r n^m = \sum_{k=0}^{m} \left[ \binom{n + \frac{k}{2} + \frac{2r-1}{2}}{k+r} - \sum_{s=0}^{r-1} \binom{\frac{k}{2} + \frac{2(r-s)-1}{2}}{k+r-s} \Sigma^s n^0 \right] \frac{\delta^{k+1} 0^{m+1}}{k+1}.$$

*Proof.* By repeatedly applied Lemma (1.8), and Equation (1). □

By Lemma (1.9), from Theorem (4.7) follows

**Proposition 4.8** (Multifold sums of powers at zero binomial form). *For non-negative integers $r, n, m$*

$$\Sigma^r n^m = \sum_{k=0}^{m} \left[ \binom{n + \frac{k}{2} + \frac{2r-1}{2}}{k+r} - \sum_{s=0}^{r-1} \binom{\frac{k}{2} + \frac{2(r-s)-1}{2}}{k+r-s} \binom{s+n-1}{s} \right] \frac{\delta^{k+1} 0^{m+1}}{k+1}.$$

*Proof.* By repeatedly applied Lemma (1.8), and Equation (1). □

Which, as well, can be expressed in terms of central factorial numbers of the second kind

$$\Sigma^r n^m = \sum_{k=0}^{m} \left[ \binom{n + \frac{k}{2} + \frac{2r-1}{2}}{k+r} - \sum_{s=0}^{r-1} \binom{\frac{k}{2} + \frac{2(r-s)-1}{2}}{k+r-s} \binom{s+n-1}{s} \right] k! T(k+1, m+1).$$

Thus, we have discussed quite remarkable results, that follow from centered Newton's formula evaluated at zero. For example,

| $n/k$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | | | | | | | | |
| 1 | 0 | 1 | | | | | | | |
| 2 | 0 | 0 | 1 | | | | | | |
| 3 | 0 | $\frac{1}{4}$ | 0 | 1 | | | | | |
| 4 | 0 | 0 | 1 | 0 | 1 | | | | |
| 5 | 0 | $\frac{1}{16}$ | 0 | $\frac{5}{2}$ | 0 | 1 | | | |
| 6 | 0 | 0 | 1 | 0 | 5 | 0 | 1 | | |
| 7 | 0 | $\frac{1}{64}$ | 0 | $\frac{91}{16}$ | 0 | $\frac{35}{4}$ | 0 | 1 | |
| 8 | 0 | 0 | 1 | 0 | 21 | 0 | 14 | 0 | 1 |

**Table 2.** Central factorial numbers of the second kind $T(n, k)$. See also the sequences A395862, and A370703 in the OEIS [11].

## 5. Conclusions

In this manuscript, we derive closed formulas for multifold sums of powers of integers by combining the central Newton interpolation formula with hockey-stick identities for binomial coefficients. We further obtain representations of multifold sums of powers in terms of Stirling numbers of the second kind and Eulerian numbers. Finally, we provide Wolfram Mathematica programs for the efficient verification of the derived identities.

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## Mathematica programs

Use this GitHub repository to validate the results using Mathematica programs.

| **Mathematica Function** | **Validates / Prints** |
|---|---|
| `ValidateOrdinarySumsOfPowers.txt` | Validates Prop. (3.3) |
| `ValidateDoubleSumsOfPowers.txt` | Validates Prop. (3.5) |
| `ValidateMultifoldSumsOfPowers.txt` | Validates Thm. (3.7) |
| `ValidateMultifoldBinomialSumsOfPowers.txt` | Validates Prop. (3.8) |
| `ValidateOrdinarySumsOfPowersAtZero.txt` | Validates Prop. (4.5) |
| `ValidateDoubleSumsOfPowersAtZero.txt` | Validates Prop. (4.6) |
| `ValidateMultifoldSumsOfPowersAtZero.txt` | Validates Thm. (4.7) |
| `ValidateMultifoldSumsOfPowersAtZeroBinomialForm.txt` | Validates Thm. (4.8) |

## Metadata


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